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Orders, Fermat and Euler

Level 4 Olympiad · Number theory

How the order of a number modulo n controls divisibility of powers, with Fermat, Euler, primitive roots and lifting the exponent.

By the end you can: You will be able to find and use the order of a number modulo n, and prove that it divides $\varphi(n)$.; You will be able to pin down the prime factors of expressions such as $a^m + 1$ using orders.; You will be able to find the exact power of a prime dividing $a^n - b^n$ with lifting the exponent..

Lesson

Does 7 ever divide 2^n+1? Write out the powers of 2 modulo 7: they run $2, 4, 1, 2, 4, 1, \ldots$, repeating every three steps, and $-1 \equiv 6$ never turns up. So $7 \nmid 2^n + 1$ for every n. Now consider $3^n + 1$. The powers of 3 modulo 7 are $3, 2, 6, 4, 5, 1$ and then repeat, and $3^3 \equiv 6 \equiv -1$, so $7 \mid 3^n + 1$ exactly when $n \equiv 3 \pmod 6$.

In general the powers of a modulo n are periodic, and the length of the period determines which exponents satisfy a given congruence. This unit studies that period.

The order of a number

Let $n \geq 2$ and let a be an integer with $\gcd(a, n) = 1$. The **order** of a modulo n, written $\operatorname{ord}_n(a)$, is the smallest positive integer d with $a^d \equiv 1 \pmod n$.

Such a d exists. There are only n residues, so two of $a^0, a^1, \ldots, a^n$ agree, say $a^i \equiv a^j$ with $i < j$. Because a is coprime to n we may cancel it i times, which leaves $a^{j-i} \equiv 1$. So some positive power is 1, and hence there is a smallest one.

> **Order Lemma.** Let $d = \operatorname{ord}_n(a)$. Then $a^k \equiv 1 \pmod n$ if and only if $d \mid k$. Consequently $a^i \equiv a^j \pmod n$ if and only if $i \equiv j \pmod d$.

Proof. If $k = dm$ then $a^k = (a^d)^m \equiv 1$. Conversely, suppose $a^k \equiv 1$ and divide: $k = qd + r$ with $0 \leq r < d$. Then $1 \equiv a^k = (a^d)^q a^r \equiv a^r$. Since $r < d$ and d is the smallest positive exponent giving 1, we must have $r = 0$. The second statement follows by cancelling the smaller power. □

Two consequences of the lemma are used constantly. First, if $a^m \equiv 1$ and $a^k \equiv 1$ then d divides both, so $a^{\gcd(m,k)} \equiv 1$. Second, to show that the order is exactly d, it is enough to check that $a^d \equiv 1$ and that $a^{d/q} \not\equiv 1$ for every prime $q \mid d$, because every proper divisor of d divides some d/q.

> **Example 1.** Find the smallest positive integer n such that 1001 divides $2^n - 1$.
>
> *Solution.*
> One could multiply by 2 modulo 1001 until 1 appears, but it is quicker to split the modulus: $1001 = 7 \cdot 11 \cdot 13$, and $1001 \mid 2^n - 1$ if and only if each of 7, 11, 13 divides $2^n - 1$. By the Order Lemma that happens if and only if n is a multiple of each of the three orders, so the answer is their lowest common multiple.
> Modulo 7, $2^3 = 8 \equiv 1$ and $2 \not\equiv 1$, so the order is 3. Modulo 11, $2^5 = 32 \equiv -1$, so $2^{10} \equiv 1$ while $2^5 \not\equiv 1$ and $2^2 = 4 \not\equiv 1$; the primes dividing 10 are 2 and 5, so by the test above the order is 10. Modulo 13, $2^6 = 64 \equiv -1$, so $2^{12} \equiv 1$, while $2^6 \not\equiv 1$ and $2^4 = 16 \equiv 3 \not\equiv 1$; the primes dividing 12 are 2 and 3, so the order is 12.

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Hence the answer is $\operatorname{lcm}(3, 10, 12) = 60$.

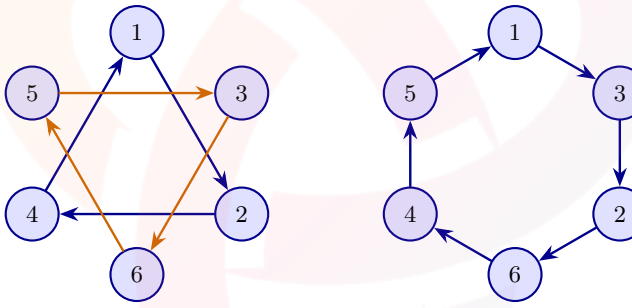
Fermat and Euler

Write $\varphi(n)$ for the number of integers in $\{1, 2, \ldots, n\}$ that are coprime to n. For a prime, $\varphi(p) = p-1$.

Euler's Theorem. If $\gcd(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod{n}$. In particular (Fermat's Little Theorem) $a^{p-1} \equiv 1 \pmod{p}$ for a prime p not dividing a. Combined with the Order Lemma: $\operatorname{ord}_n(a) \mid \varphi(n)$, and $\operatorname{ord}_p(a) \mid p-1$.

Proof. Let $r_1, \ldots, r_k$ be the residues coprime to n, where $k = \varphi(n)$. The numbers $ar_1, \ldots, ar_k$ are also coprime to n, and no two are congruent, since $ar_i \equiv ar_j$ gives $r_i \equiv r_j$ after cancelling a. So they are the same k residues in a different order, and multiplying everything together gives $a^k r_1 \cdots r_k \equiv r_1 \cdots r_k$. The product $r_1 \cdots r_k$ is coprime to n, so it cancels. □

For a prime, a second proof explains why the order divides $p-1$. Multiplication by a permutes the nonzero residues, and the permutation splits into cycles. The cycle through r is $r, ar, a^2r, \ldots$, and it first returns to r after exactly $\operatorname{ord}_p(a)$ steps, since $a^k r \equiv r$ means $a^k \equiv 1$. So every cycle has length $\operatorname{ord}_p(a)$, and since the cycles partition the $p-1$ nonzero residues, $\operatorname{ord}_p(a) \mid p-1$.



$x \mapsto 2x \pmod 7$ two cycles of length 3 $x \mapsto 3x \pmod 7$ one cycle of length 6

Orders most often enter olympiad problems through expressions like $a^m + 1$. If an odd prime p divides $a^m + 1$, then $a^{2m} \equiv 1$ but $a^m \equiv -1 \not\equiv 1$, so the order divides $2m$ but does not divide m.

Prime factors of $a^{2^k} + 1$. If an odd prime p divides $a^{2^k} + 1$, then $\operatorname{ord}_p(a) = 2^{k+1}$, and therefore $p \equiv 1 \pmod{2^{k+1}}$.

Proof. The order divides 2^{k+1}, so it is 2^j for some $j \le k+1$. If $j \le k$ then $a^{2^k} = (a^{2^j})^{2^{k-j}} \equiv 1$, contradicting $a^{2^k} \equiv -1$ (and $1 \not\equiv -1$ because p is odd). So $j = k+1$, and $2^{k+1} \mid p-1$. □

Hence every odd prime factor of $n^2 + 1$ is 1 modulo 4, and every odd prime factor of $n^4 + 1$ is 1 modulo 8.

Example 2. Find all pairs of primes (p, q) such that $p \mid 2^q + 1$ and $q \mid 2^p + 1$.

Solution.
Small cases suggest that $p = q = 3$ is the only answer. To relate p and q we use the order of 2 modulo each prime.
Both $2^q + 1$ and $2^p + 1$ are odd, so p and q are odd. From $p \mid 2^q + 1$ we get $2^{2q} \equiv 1 \pmod{p}$ while $2^q \equiv -1 \not\equiv 1$. Let $d = \operatorname{ord}_p(2)$. Then $d \mid 2q$ and $d \nmid q$. The divisors of $2q$ are $1, 2, q, 2q$,

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and the condition $d \nmid q$ removes 1 and q. If $d = 2$ then $p \mid 2^2 - 1 = 3$, so $p = 3$. If $d = 2q$ then $2q \mid p - 1$ by Fermat, so $p > q$.
So either $p = 3$ or $p > q$. By symmetry, either $q = 3$ or $q > p$. We cannot have both $p > q$ and $q > p$, so at least one of the primes is 3. If $p = 3$ then $q \mid 2^3 + 1 = 9$, so $q = 3$; if $q = 3$ then likewise $p = 3$. The only pair is $(3, 3)$.

Example 3. Let n be an integer. Prove that every prime factor p of $n^4 - n^2 + 1$ satisfies $p \equiv 1 \pmod{12}$.

Solution.
The expression is not of the form $a^m + 1$, so the lemma above does not apply directly. Instead we use the factorisation $(n^2 + 1)(n^4 - n^2 + 1) = n^6 + 1$. So $p \mid n^6 + 1$, giving $n^6 \equiv -1$ and $n^{12} \equiv 1 \pmod{p}$.
The number $n^4 - n^2 + 1 = n^2(n^2 - 1) + 1$ is odd, so p is odd. Also $p \nmid n$, otherwise $n^4 - n^2 + 1 \equiv 1$. Let $d = \mathrm{ord}_p(n)$. Then $d \mid 12$ and $d \nmid 6$, which leaves $d = 4$ or $d = 12$.
The case $d = 4$ has to be ruled out separately, and many solutions skip it. Suppose $d = 4$. Then $n^4 \equiv 1$, so $0 \equiv n^4 - n^2 + 1 \equiv 2 - n^2$, giving $n^2 \equiv 2$ and so $n^4 \equiv 4$. Together with $n^4 \equiv 1$ this forces $p \mid 3$. But 3 never divides $n^4 - n^2 + 1$: if $3 \mid n$ the value is 1 modulo 3, and otherwise $n^2 \equiv 1$ and the value is $1 - 1 + 1 = 1$ modulo 3. So $d = 4$ is impossible, $d = 12$, and $12 \mid p - 1$. □

Primitive roots, and why $x^d \equiv 1$ has few solutions

Lagrange's Theorem. Let p be prime and f a polynomial with integer coefficients and degree $d \geq 1$, whose leading coefficient is not divisible by p. Then $f(x) \equiv 0 \pmod{p}$ has at most d solutions modulo p.

Proof. Induct on d; the case $d = 1$ is clear because the leading coefficient is invertible. If r is a root, polynomial division gives $f(x) = (x - r)g(x) + f(r)$ with g of degree $d - 1$, integer coefficients and the same leading coefficient. If $s \not\equiv r$ is another root, then $(s - r)g(s) \equiv 0$, and since p is prime and $p \nmid s - r$, we get $g(s) \equiv 0$. So the other roots are roots of g, and there are at most $d - 1$ of them. □

Note that the modulus must be prime: $x^2 \equiv 1 \pmod{8}$ has four solutions. A consequence is that k-th powers take few values modulo p (see L4-NT2).

A harder fact, which we state without proof, is the **Primitive Root Theorem**: for every prime p there is a residue g with $\mathrm{ord}_p(g) = p - 1$. Then $g, g^2, \ldots, g^{p-1}$ run through all the nonzero residues, and for every $d \mid p - 1$ the residue $g^{(p-1)/d}$ has order exactly d. So *an element of order d exists modulo p if and only if $d \mid p - 1$*. The proof counts elements of each order using Lagrange's Theorem; it is not hard, and is left as an exercise for the reader.

Lifting the exponent

Orders tell you when p divides $a^n - b^n$. Lifting the exponent gives the exact power of p that divides it. Write $v_p(m)$ for the exponent of p in m.

Lifting the Exponent (LTE). Let p be an odd prime and let a, b be integers with $p \mid a - b$ and $p \nmid ab$. Then for every positive integer n,

$$v_p(a^n - b^n) = v_p(a - b) + v_p(n).$$

For $p = 2$: if a, b are odd and n is even, then $v_2(a^n - b^n) = v_2(a - b) + v_2(a + b) + v_2(n) - 1$.

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Proof for odd p. Step 1: if $p \nmid n$, write $a^n - b^n = (a-b)S$ with $S = a^{n-1} + a^{n-2}b + \cdots + b^{n-1}$. As $a \equiv b$, each of the n terms of S is congruent to a^{n-1}, so $S \equiv na^{n-1} \not\equiv 0 \pmod p$, and $v_p(a^n - b^n) = v_p(a-b)$.

Step 2: the case $n = p$. Put $a = b + t$ with $p \mid t$. By the binomial theorem $(b+t)^m \equiv b^m + mtb^{m-1} \pmod{t^2}$, and $p^2 \mid t^2$, so modulo p^2

$$S = \sum_{i=0}^{p-1} (b+t)^{p-1-i} b^i \equiv pb^{p-1} + tb^{p-2} \sum_{i=0}^{p-1} (p-1-i) = pb^{p-1} + tb^{p-2} \cdot p \cdot \frac{p-1}{2}.$$

Because p is odd, $\frac{p-1}{2}$ is an integer and the last term is divisible by pt, hence by p^2. So $S \equiv pb^{p-1} \pmod{p^2}$, giving $v_p(S) = 1$ and $v_p(a^p - b^p) = v_p(a-b) + 1$.

For general $n = p^k m$ with $p \nmid m$, apply Step 2 to a, b, then to a^p, b^p, and so on k times (the hypotheses survive each time), and finish with Step 1. □

For $p = 2$, write $n = 2^k m$ with m odd. The odd part changes nothing, by the argument of Step 1 (a sum of m odd terms is odd), and $a^{2^k} - b^{2^k} = (a-b)(a+b)(a^2+b^2)(a^4+b^4)\cdots(a^{2^{k-1}} + b^{2^{k-1}})$, where each factor $a^{2^j} + b^{2^j}$ with $j \geq 1$ is a sum of two odd squares, so it is 2 modulo 4. That gives the extra $k - 1$.

Example 4. Prove that for every positive integer k, the order of 2 modulo 3^k is $2 \cdot 3^{k-1}$.

Solution.
We need to know which n make $3^k \mid 2^n - 1$. LTE counts powers of 3, but it needs $3 \mid a - b$, and $3 \nmid 2 - 1$. So we first find the order of 2 modulo 3: since $2 \equiv -1$, we have $2^n \equiv 1 \pmod 3$ if and only if n is even.
So if $3^k \mid 2^n - 1$ then $n = 2m$, and $2^n - 1 = 4^m - 1$ where now $3 \mid 4 - 1$. LTE gives $v_3(4^m - 1) = v_3(3) + v_3(m) = 1 + v_3(n)$. Therefore $3^k \mid 2^n - 1$ if and only if n is even and $v_3(n) \geq k - 1$, that is, if and only if $2 \cdot 3^{k-1} \mid n$. The smallest such positive n is $2 \cdot 3^{k-1}$, which is the order. □
So 2 is a primitive root modulo every power of 3. The pattern "find the order modulo p, then lift with LTE" is the standard method for AIME questions asking for the least n with $p^k \mid a^n - b^n$.

Common traps.

- The order only exists when $\gcd(a, n) = 1$. Check this before you write $\mathrm{ord}_n(a)$.
- The order divides $p - 1$ but is usually smaller. Never assume 2 has order $p - 1$ without checking.
- From $a^m \equiv -1$ you learn that the order divides $2m$ but not m. List all such divisors, including the small ones: the cases $d = 2$ in Example 2 and $d = 4$ in Example 3 are where marks are most often lost.
- LTE needs $p \mid a - b$ and $p \nmid ab$. If $p \nmid a - b$, first raise to the order. Using the odd-prime formula at $p = 2$ is wrong: $v_2(3^2 - 1) = 3$, not $v_2(2) + v_2(2) = 2$.
- Lagrange's bound needs a prime modulus.

How to spot this in a contest

Think orders whenever a problem involves $a^n \pm 1$, $a^n - b^n$ or "for which n does p divide": the answer is usually a congruence class modulo an order. Prime factors of $n^2 + 1$, $n^4 + 1$ or anything dividing $n^k \pm 1$ point to the "order divides $2m$ but not m" argument. "Largest power of p dividing" or "least n with $p^k \mid \ldots$" is LTE, after first reaching the order modulo p. Exponent towers are reduced by working modulo the order (or φ) in the exponent.

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Finally, a standard opening (the smallest prime factor trick) for problems of the form $n \mid a^n - b^n$ or $n \mid a^n + 1$: look at the **smallest prime factor** p of n. The order of a (or of ab^{-1}) modulo p divides both something built from n and $p-1$, and since every prime factor of n is at least p while every prime factor of $p-1$ is less than p, $\gcd(n, p-1) = 1$, so the order is usually forced to divide 2. Practice problem 6 uses this argument.

Practice

Try each problem before turning to the hints and solutions at the back.

1. $\star$ Find the order of 13 modulo 100, and use it to find the last two digits of 13^{2026}.

2. $\star\star$ Find all positive integers n such that 13 divides $2^n + 3^n$.

3. $\star\star$ Find the largest integer k such that 10^k divides $3^{2000} - 1$.

4. $\star\star\star$ Find the sum of all primes $p < 200$ for which there is an integer n with $p \mid n^6 + n^3 + 1$. (You may use the Primitive Root Theorem.)

5. $\star\star\star$ Prove that there are infinitely many primes of the form $8k + 1$.

6. $\star\star\star\star$ Let n be an odd positive integer such that n divides $3^n + 1$. Prove that $n = 1$.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. UKMT BMO1 2005/06 Problem 1 $\star$
2. AIME 1983 Problem 6 $\star$
3. UKMT BMO1 2006/07 Problem 1 $\star\star$
4. AIME I 2011 Problem 11 $\star\star$
5. UKMT BMO1 2013/14 Problem 3 $\star\star\star$
6. AIME I 2020 Problem 12 $\star\star\star$
7. AIME I 2019 Problem 14 $\star\star\star$
8. AIME II 2021 Problem 13 $\star\star\star$
9. UKMT BMO1 2024/25 Problem 5 $\star\star\star\star$
10. AIME I 2024 Problem 13 $\star\star\star\star$
11. AIME II 2023 Problem 15 $\star\star\star\star$

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Hints

1. Work modulo 4 and modulo 25 separately; the order modulo 100 is the lowest common multiple of the two orders.
2. Divide through by 3^n: the condition is about a single number raised to the power n.
3. Find the powers of 2 and of 5 separately. For 5, you need to reach the order of 3 modulo 5 before LTE applies.
4. Multiply by $n^3 - 1$. What can the order of n modulo p be?
5. Suppose there are only finitely many and build a number of the form $N^4 + 1$ from them. What do its prime factors look like?
6. Suppose $n > 1$ and let p be the smallest prime factor of n. The order of 3 modulo p divides two numbers with very little in common.

Solutions

Problem 1

Answer: The order is 20; the last two digits are 09.

Modulo 4 we have $13 \equiv 1$, so the order there is 1. Modulo 25 the order divides $\varphi(25) = 20$. Now $13^2 = 169 \equiv 19 \equiv -6$, so $13^4 \equiv 36 \equiv 11$ and $13^5 \equiv 143 \equiv -7$, giving $13^{10} \equiv 49 \equiv -1 \pmod{25}$. Hence $13^{20} \equiv 1$, while $13^{10} \not\equiv 1$ and $13^4 \not\equiv 1$. Every proper divisor of 20 divides 10 or 4, so the order modulo 25 is 20.

A power 13^n is 1 modulo 100 exactly when it is 1 modulo both 4 and 25, that is, when $20 \mid n$. So $\operatorname{ord}_{100}(13) = 20$. Since $2026 = 20 \cdot 101 + 6$, the Order Lemma gives $13^{2026} \equiv 13^6 \pmod{100}$. From $13^2 \equiv 69$ we get $13^4 \equiv 69^2 = 4761 \equiv 61$ and $13^6 \equiv 61 \cdot 69 = 4209 \equiv 9$. The last two digits are 09.

Problem 2

Answer: Exactly the n with $n \equiv 2 \pmod{4}$.

Since $3 \cdot 9 = 27 \equiv 1 \pmod{13}$, the inverse of 3 is 9. Multiplying by 9^n, the condition $2^n + 3^n \equiv 0$ becomes $(2 \cdot 9)^n \equiv -1$, that is, $5^n \equiv -1 \pmod{13}$.

Note that $5^2 = 25 \equiv -1$, so $5^4 \equiv 1$ and the order of 5 is 4. The powers of 5 cycle through $5, -1, -5, 1$, so $5^n \equiv -1$ exactly when $n \equiv 2 \pmod{4}$. These are the required n; for example $2^2 + 3^2 = 13$.

Problem 3

Answer: 4

For the prime 2: 3 and 1 are odd and 2000 is even, so the $p = 2$ form of LTE gives $v_2(3^{2000} - 1) = v_2(3 - 1) + v_2(3 + 1) + v_2(2000) - 1 = 1 + 2 + 4 - 1 = 6$.

For the prime 5: LTE cannot be applied to $3^{2000} - 1^{2000}$ directly, because $5 \nmid 3 - 1$. But $3^2 = 9 \equiv -1 \pmod{5}$, so $\operatorname{ord}_5(3) = 4$, and $3^{2000} - 1 = 81^{500} - 1$ with $5 \mid 81 - 1$. LTE now gives $v_5(81^{500} - 1) = v_5(80) + v_5(500) = 1 + 3 = 4$.

So $3^{2000} - 1$ is divisible by 2^6 and by 5^4 but not by 5^5, and the largest power of 10 dividing it is 10^4. Thus $k = 4$.

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Problem 4

Answer: 911

Note first that $n^6 + n^3 + 1$ is always odd, since $n^6 + n^3 = n^3(n^3 + 1)$ is even; so $p \neq 2$. Also $(n^3 - 1)(n^6 + n^3 + 1) = n^9 - 1$.

Suppose $p \mid n^6 + n^3 + 1$. Then $p \nmid n$ and $n^9 \equiv 1 \pmod{p}$, so $d = \operatorname{ord}_p(n)$ divides 9. If $d \mid 3$ then $n^3 \equiv 1$, so $n^6 + n^3 + 1 \equiv 3$ and $p = 3$. Otherwise $d = 9$, and $9 \mid p - 1$ by Fermat's Little Theorem.

Conversely, $p = 3$ works with $n = 1$, since $1 + 1 + 1 = 3$. If $p \equiv 1 \pmod{9}$, the Primitive Root Theorem gives an n of order 9 modulo p. Then $p \mid n^9 - 1 = (n^3 - 1)(n^6 + n^3 + 1)$, while $p \nmid n^3 - 1$ because the order is not a divisor of 3. Hence $p \mid n^6 + n^3 + 1$.

So the primes are 3 together with the primes below 200 that are 1 modulo 9. Running through $10, 19, 28, \ldots, 199$, these are $19, 37, 73, 109, 127, 163, 181, 199$ (the others, such as $91 = 7 \cdot 13$ and $145 = 5 \cdot 29$, are composite or even). The sum is $3 + 19 + 37 + 73 + 109 + 127 + 163 + 181 + 199 = 911$.

Problem 5

Suppose, for contradiction, that there are only finitely many primes congruent to 1 modulo 8; call them $p_1, \ldots, p_r$ (possibly $r = 0$). Let $N = 2p_1p_2 \cdots p_r$, with $N = 2$ if $r = 0$, and let p be any prime factor of $N^4 + 1$.

Since N is even, $N^4 + 1$ is odd, so p is odd. Also $p \nmid N$, since otherwise $p \mid 1$. From $N^4 \equiv -1 \pmod{p}$ we get $N^8 \equiv 1$, so $d = \operatorname{ord}_p(N)$ divides 8. If d divided 4 we would have $N^4 \equiv 1$, hence $1 \equiv -1$ and $p \mid 2$, which is false. Therefore $d = 8$. By Fermat's Little Theorem and the Order Lemma, $8 \mid p - 1$, so $p \equiv 1 \pmod{8}$.

So p is one of $p_1, \ldots, p_r$, and hence $p \mid N$. This contradicts $p \nmid N$. Therefore there are infinitely many primes of the form $8k + 1$. □

Problem 6

Suppose, for contradiction, that $n > 1$, and let p be the smallest prime factor of n. Since n is odd, p is odd. Also $p \neq 3$, because $3^n + 1 \equiv 1 \pmod{3}$ and p divides $3^n + 1$.

From $p \mid 3^n + 1$ we get $3^n \equiv -1$ and so $3^{2n} \equiv 1 \pmod{p}$. Let $d = \operatorname{ord}_p(3)$, which exists as $p \neq 3$. By the Order Lemma $d \mid 2n$, and by Fermat's Little Theorem $d \mid p - 1$. Hence $d \mid \gcd(2n, p - 1)$.

We claim $\gcd(2n, p - 1) = 2$. Let q be an odd prime dividing this gcd. Then $q \mid n$, so $q \geq p$ by the choice of p; but also $q \mid p - 1$, so $q \leq p - 1$. This is impossible, so the gcd is a power of 2. As n is odd, $4 \nmid 2n$, so the gcd divides 2; and it equals 2 since $p - 1$ is even.

Therefore $d \mid 2$, so $3^2 \equiv 1 \pmod{p}$, that is, $p \mid 8$. This contradicts p being odd. Hence $n = 1$, which does satisfy $1 \mid 3 + 1$. □

Remark. The oddness of n is essential: $n = 2$ and $n = 10$ both divide $3^n + 1$, since $3^{10} + 1 = 59050$.

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